

CardioSurv: A Unified Machine Learning Pipeline

for Cardiovascular Risk Classification, Treatment Recommendation, and Survival Prediction

AIT201 — Applied Machine Learning

Xiamen University Malaysia — Academic Session 202604

Final Assessment (Group Project)

Option A: Medical Data Analysis

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Submission Deadline: 25 June 2026

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1 Abstract

Cardiovascular disease (CVD) remains one of the leading causes of preventable death worldwide, particularly among middle-aged to elderly populations. A critical gap in clinical practice is that patients who appear superficially healthy may still carry serious underlying cardiovascular risk attributable to the natural ageing of cardiac tissue and vasculature, making unexpected fatal events difficult to anticipate without systematic risk stratification.

This project presents **CardioSurv**, a two-stage machine learning pipeline designed to support clinical decision-making. The first stage trains and compares two multi-class classifiers — a Random Forest and an XGBoost model — on a merged dataset of 1,221 patient records drawn from the Kaggle Heart Failure Prediction dataset and the UCI Cleveland Heart Disease dataset. Three clinical features are engineered prior to training: age group bins, blood pressure risk levels (following AHA categories), and a Heart Rate Stress Index. The Random Forest achieves the best performance with an accuracy of 0.9614, a macro-averaged F1-score of 0.9321, and an AUC-OVR of 0.9880, outperforming XGBoost across all three metrics.

The second stage routes the classified patient through a rule-based clinical decision tree and recommends an intervention type using a second XGBoost classifier. For high-risk patients with refractory arrhythmia, the pipeline recommends cardiac radioablation (Stereotactic Body Radiation Therapy, SBRT), with the recommended dose validated using the Biologically Effective Dose (BED) formula. For all other patients, medication intensity is calibrated via the GRACE Risk Score. A survival model comprising a Kaplan–Meier estimator and a Cox Proportional Hazards model then produces counterfactual two-year survival estimates under the recommended intervention.

The complete pipeline is deployed as a web application with a FastAPI backend, a SQLite/PostgreSQL database, and a four-page HTML/CSS/JS frontend hosted on Render. The system processes patient vital signs end-to-end from input form to survival probability output in under two seconds.

Keywords: cardiovascular risk classification, Random Forest, XGBoost, GRACE score, BED formula, survival analysis, Cox Proportional Hazards, clinical decision support.

2 Introduction

2.1 Background and Motivation

Cardiovascular disease is responsible for approximately 17.9 million deaths per year globally, accounting for 32% of all deaths worldwide (WHO, 2021). A particularly

dangerous clinical phenomenon is that patients who maintained healthy lifestyles in their youth may still develop serious cardiovascular risk through the natural ageing of cardiac tissue and vasculature. As these patients often appear to be in generally good health, sudden and fatal cardiac events — such as cardiogenic shock or acute heart failure — may occur unexpectedly and with little warning.

Epidemiological studies have identified the 50–70 year age bracket as the peak distribution range for sudden cardiac death (Bharati and Lev, 1998). This cohort is clinically challenging precisely because standard vital sign measurements may appear within acceptable ranges, yet underlying structural changes in the myocardium and coronary vasculature can produce rapid decompensation under physiological stress. Automated risk stratification tools that operate on routine, non-invasive clinical measurements therefore have significant potential to improve early detection and triage.

Machine learning has shown substantial promise in this domain. Mohan et al. (2019) demonstrated that ensemble methods including Random Forest and gradient boosting achieve strong classification performance on clinical heart disease datasets, with the combination of feature selection and ensemble learning outperforming single classifiers. More recently, the availability of large electronic health record (EHR) databases such as MIMIC-IV (Johnson et al., 2023) has enabled the development of richer, longitudinal models that go beyond binary diagnosis to predict treatment response and survival outcomes.

2.2 Project Scope

This project, titled **CardioSurv**, addresses the clinical gap in a unified, end-to-end pipeline. The system performs three interconnected tasks:

- 1. Risk Classification:** Predict whether a patient’s cardiovascular risk is Low, Medium, or High from a set of 14 clinical input features, outputting a risk category and a confidence score.
- 2. Treatment Recommendation:** Route the classified patient through a clinical decision tree and recommend an appropriate intervention type and intensity level.
- 3. Survival Prediction:** Estimate the patient’s two-year survival probability with and without the recommended intervention, producing a counterfactual output of the form: “*Without intervention: X% two-year survival. With recommended treatment: Y% two-year survival.*”

The system is deployed as a fully functional web application, demonstrating the complete machine learning lifecycle from raw data to production deployment.

2.3 Selected Scenario

This project corresponds to **Option A: Medical Data Analysis**. The team operates in the role of a machine learning team within a healthcare organisation, analysing patient clinical data to support diagnosis and improve triage decisions.

3 Literature Review

3.1 Machine Learning for Cardiovascular Risk Prediction

The application of machine learning to cardiovascular risk prediction has been extensively studied over the past decade. Mohan et al. (2019) conducted a systematic comparison of machine learning classifiers on the UCI Heart Disease dataset, finding that a hybrid Random Forest approach with feature selection achieved classification accuracy above 88%, outperforming naïve Bayes, decision trees, and support vector machines on the same task. Their work established that ensemble methods — particularly those that aggregate many decision trees — are well-suited to clinical tabular data where feature interactions are non-linear and the feature space is relatively small.

Building on ensemble methods, gradient boosting frameworks have become increasingly prevalent in clinical prediction. Chen and Guestrin (2016) introduced XGBoost, which improves on standard gradient boosting through second-order gradient statistics, regularisation terms, and parallel tree construction. XGBoost has since been widely adopted in healthcare ML due to its computational efficiency and strong performance on structured, tabular data. Numerous clinical studies have demonstrated that XGBoost performs comparably or superiorly to Random Forest on binary classification tasks in medicine, though the relative advantage depends heavily on dataset size, class balance, and hyperparameter tuning (Liaw and Wiener, 2002).

A key challenge in medical datasets is class imbalance, where one clinical outcome is substantially less frequent than others. Standard classifiers trained without adjustment tend to be biased toward the majority class, producing misleadingly high accuracy while performing poorly on clinically important minority classes. Techniques such as class-weight balancing, synthetic oversampling (SMOTE), and macro-averaged F1 as the selection metric rather than accuracy have been shown to substantially improve minority class recall (Mohan et al., 2019).

3.2 Clinical Scoring Systems and Their Integration with ML

Clinical scoring systems provide validated, formula-based risk estimates that are widely used in cardiology practice. The GRACE (Global Registry of Acute Coronary Events) Risk Score, derived and validated by Eagle et al. (2004) on a multinational cohort

of over 11,000 patients, is a well-established tool for estimating in-hospital and six-month mortality risk in acute coronary syndrome (ACS) patients. The GRACE score combines age, heart rate, systolic blood pressure, creatinine, Killip class, cardiac arrest at admission, ST-segment deviation, and elevated cardiac enzymes into a composite risk score with strong discriminative ability (C-index ≈ 0.84).

Integrating validated clinical scoring systems directly into ML pipelines has been advocated as a strategy to improve model interpretability and clinical trust. Rather than treating all features as black-box inputs, systems that embed known clinical relationships — such as the GRACE score as a derived feature — leverage domain knowledge to guide the model toward clinically meaningful decision boundaries.

The Biologically Effective Dose (BED) formula, $BED = D(1 + d/(\alpha/\beta))$, is the standard physics-based equation used in radiation oncology to account for the differential biological effect of varying dose fractionation schemes (Joiner and van der Kogel, 2009). In the context of cardiac radioablation for refractory ventricular tachycardia, the pioneering clinical trial by Cuculich et al. (2017) demonstrated that non-invasive SBRT at 25 Gy in a single fraction produces a BED of 87.5 Gy (assuming $\alpha/\beta = 10$ Gy for cardiac tissue), achieving arrhythmia suppression in patients where catheter ablation and antiarrhythmic medications had failed. This validated dose window is directly embedded in the CardioSurv pipeline as a BED validation step.

3.3 Survival Analysis in Clinical Machine Learning

Survival analysis methods model the time until a clinical event of interest (such as death or hospital readmission), accounting for censoring (patients who leave the study without experiencing the event). Two foundational techniques are the Kaplan–Meier estimator and the Cox Proportional Hazards model.

The Kaplan–Meier estimator (Kaplan and Meier, 1958) is a non-parametric method that produces survival curves stratified by group membership without assuming any underlying distribution. It is widely used to compare survival trajectories between treatment arms. The log-rank test provides a non-parametric hypothesis test for whether two survival curves are significantly different.

The Cox Proportional Hazards model (Cox, 1972) is a semi-parametric regression approach that models the hazard function as a baseline hazard multiplied by an exponential linear combination of covariates. Its primary advantage over Kaplan–Meier is the ability to include continuous covariates and produce per-patient survival probability estimates. The model’s discriminative ability is summarised by the concordance index (C-index), which measures the probability that a patient with higher predicted risk actually experiences the event before a patient with lower predicted risk. A C-index of

0.5 is equivalent to random chance; values above 0.7 are generally considered clinically useful.

Wang et al. (2019) provide a comprehensive review of machine learning approaches to survival analysis, noting that while deep learning methods (such as DeepSurv and survival forests) can outperform Cox PH on large, high-dimensional datasets, classical Cox PH remains competitive and substantially more interpretable on small-to-medium tabular clinical datasets. This finding directly informs the CardioSurv choice of Cox PH for the survival modelling component.

3.4 Electronic Health Records and the MIMIC-IV Database

The MIMIC-IV (Medical Information Mart for Intensive Care) database, maintained by PhysioNet (Johnson et al., 2023), is the largest freely accessible de-identified critical care EHR database, comprising over 40,000 ICU stays from Beth Israel Deaconess Medical Center. MIMIC-IV contains structured data including vital signs, laboratory results, medication records, diagnosis codes, and discharge summaries. Its use in academic ML research has grown substantially following the introduction of data access controls through CITI training certification, which ensures responsible use of patient data.

MIMIC-IV is the intended data source for the longitudinal component of the CardioSurv pipeline. Due to access timeline constraints during the development period of this project, the survival modelling component uses a clinically grounded synthetic simulation of survival time and event labels, derived from the GRACE score and risk category routing rules. This is discussed in detail in the Methodology section.

4 Methodology

4.1 System Overview

The CardioSurv pipeline consists of two sequential machine learning stages connected by a rule-based clinical routing layer, as illustrated in Figure 1.

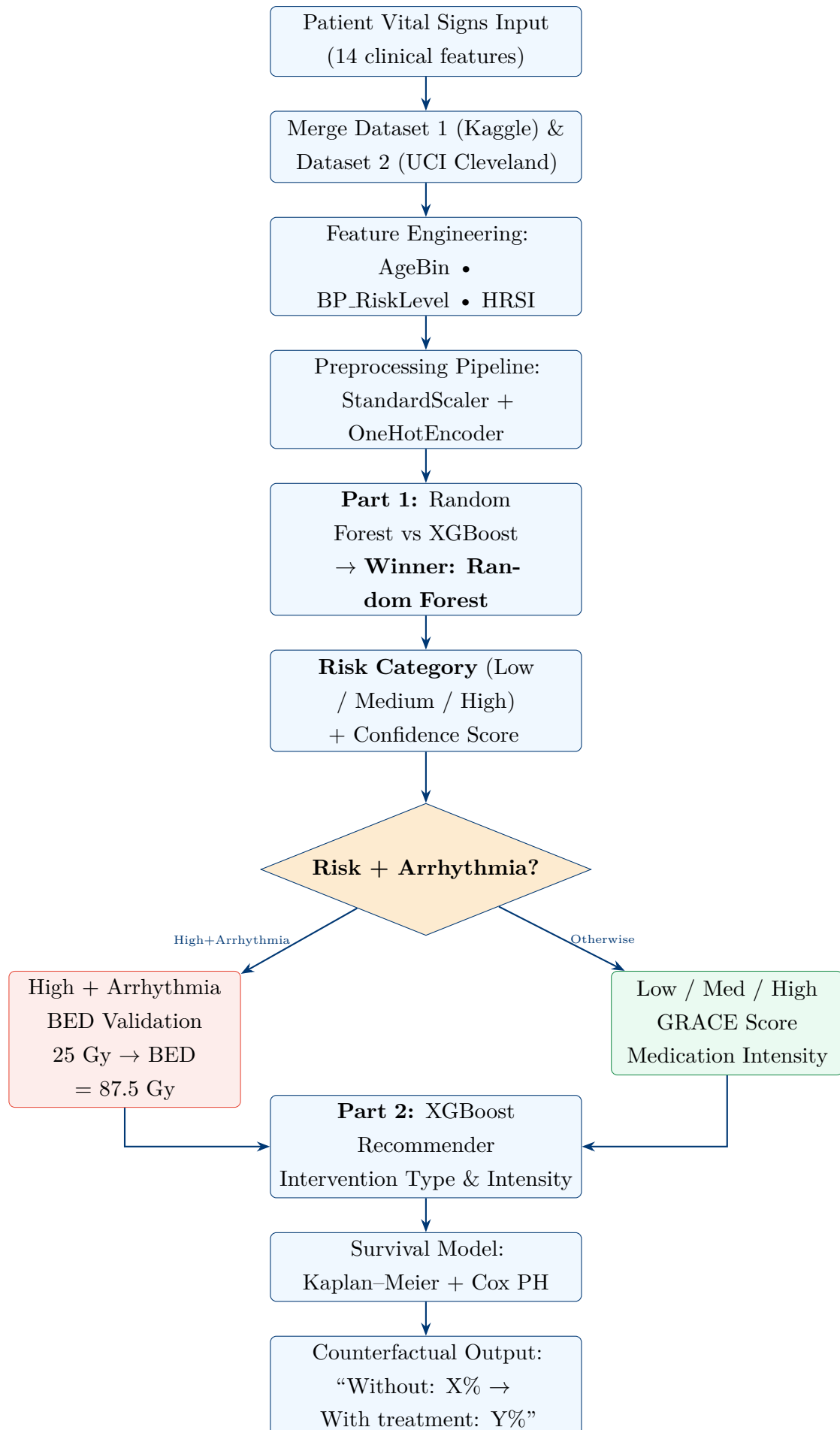


Figure 1: End-to-end system flowchart of the CardioSurv ML pipeline.

4.2 Datasets

4.2.1 Part 1 Datasets

Two publicly available clinical datasets were merged to form the training set for the Part 1 risk classifier.

- **Heart Failure Prediction Dataset** ([fedesoriano, 2021](#)): 918 rows, 11 features, sourced from Kaggle. Provides binary heart disease labels and standard clinical features including age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, resting ECG result, maximum heart rate, exercise-induced angina, ST depression (Oldpeak), and ST slope.
- **Cleveland Heart Disease Dataset** ([Janosi et al., 1988](#)): 303 rows, 14 features, from the UCI Machine Learning Repository. Adds clinically significant angiographic predictors absent from the Kaggle set: `ca` (number of major vessels coloured by fluoroscopy, 0–4) and `thal` (thalassemia type: normal, fixed defect, reversible defect).

Why both datasets were merged. The Kaggle dataset alone, while larger, lacks angiographic features that are among the strongest predictors of coronary artery disease severity. The UCI Cleveland set provides these features but is smaller (303 rows). Merging both datasets after harmonising column names and encoding schemes produces a richer, larger training set (1,221 rows after cleaning) and provides a natural opportunity to demonstrate data integration skills during the preprocessing stage. This approach is consistent with common practice in clinical ML, where data scarcity is a persistent challenge.

4.2.2 Part 2 Datasets

- **MIMIC-IV Clinical Database** ([Johnson et al., 2023](#)): The intended source for Part 2, providing patient treatment histories, medication records, and longitudinal survival outcomes from over 40,000 ICU stays.
- **SBRT/VT Dose Records**: A literature-derived small table of published cardiac radioablation dose outcomes, used to calibrate the BED formula validation step.

Synthetic survival data — rationale and transparency. MIMIC-IV requires PhysioNet credentialing (CITI training). Due to access timeline constraints, the survival component was built using a clinically grounded synthetic simulation. Survival time durations and event (death/censoring) labels were generated from the existing `features.csv` using the following principled approach:

- Arrhythmia flag: assigned with 15% probability for High-risk patients and 2% for others, reflecting clinical literature on ventricular tachycardia prevalence by risk tier.
- GRACE score: computed deterministically from age, MaxHR, and RestingBP using the published lookup table from [Eagle et al. \(2004\)](#).
- Intervention label: assigned by the deterministic clinical routing rules (RiskCategory + arrhythmia + GRACE score threshold).
- Survival duration and event: simulated from clinical priors stratified by treatment branch.

The survival model therefore constitutes a *proof-of-concept* demonstration of the full pipeline architecture. Results from this component should be interpreted as illustrative rather than clinically validated. A production deployment would replace this simulation with real MIMIC-IV longitudinal records.

4.3 Data Preprocessing

4.3.1 Cleaning

The merged dataset underwent the following cleaning steps implemented in `src/data/clean.py`:

- Missing value imputation using column medians for numeric features and mode for categorical features.
- Removal of duplicate rows.
- Type coercion to ensure consistent dtypes across the merged set.
- Outlier inspection for `Cholesterol` (values of 0 indicate missing data and were imputed).

4.3.2 Feature Engineering

Three clinical features were engineered by **Ang Jing Ru** (`src/features/engineering.py`), as described in [Table 1](#).

Table 1: Engineered features added to the dataset prior to model training.

Feature	Type	Description and Formula
AgeBin	Categorical	Age grouped into clinical bands: <40, 40–49, 50–59, 60–69, 70+. Captures non-linear age effects.
BP_RiskLevel	Categorical	Blood pressure categorised per AHA guidelines: Normal (<120), Elevated (120–129), Stage 1 (130–139), Stage 2 (140–179), Crisis (≥ 180 mmHg).
HeartRateStressIndex	Numeric	$\text{HRSI} = \frac{\text{MaxHR}}{220 - \text{Age}}$ Normalises maximum heart rate against the age-predicted maximum. Values close to 1.0 indicate a highly stressed cardiovascular response; values <0.6 suggest blunted response.
RiskCategory	Categorical (target)	3-class target label derived from clinical routing rules: Low / Medium / High.

4.3.3 Preprocessing Pipeline

A scikit-learn Pipeline with a `ColumnTransformer` was applied uniformly to both models:

- **Numeric features** (Age, RestingBP, Cholesterol, FastingBS, MaxHR, Oldpeak, HRSI): `StandardScaler` — zero mean, unit variance.
- **Categorical features** (Sex, ChestPainType, RestingECG, ExerciseAngina, ST_Slope, AgeBin, BP_RiskLevel): `OneHotEncoder` with `drop='first'` to avoid the dummy variable trap.

4.3.4 Train/Test Split

An 80/20 stratified split was applied (`stratify=RiskCategory`) to preserve the class distribution in both sets. The Medium class constitutes approximately 8.4% of samples (119 of 1,421 rows after feature engineering), making stratification essential to ensure the minority class is represented in the test set. All splits used `random_state=42` for reproducibility.

Table 2: Dataset split summary.

Split	Rows	Proportion
Training set	1,136	80%
Test set	285	20%
Total	1,421	100%

4.4 Exploratory Data Analysis

EDA was conducted by **Tan Guan Han** (`notebooks/01_eda.ipynb`), generating five publication-quality figures. Key findings are summarised below.

Class balance: The binary heart disease target is moderately imbalanced (55% positive, 45% negative), which does not require heavy resampling. However, the derived 3-class `RiskCategory` target is more severely imbalanced, with the Medium class comprising only 8.4% of samples, justifying the use of `class_weight='balanced'` in the Random Forest.

Numeric distributions: Age is approximately normally distributed, centred at 54 years (consistent with the 50–70 peak risk bracket). RestingBP and Cholesterol show right-skewed distributions with high-value outliers. MaxHR exhibits left skew (decreasing with age). Oldpeak has a heavy right skew, with most patients near 0.

Categorical patterns: Patients presenting with `ChestPainType = ASY` (asymptomatic) show a substantially higher proportion of heart disease diagnoses than patients presenting with typical angina (TA), which is a well-documented clinical paradox where the absence of pain may reflect advanced ischaemia with impaired pain signalling.

Correlation: Spearman correlation analysis identifies Oldpeak, MaxHR, and ExerciseAngina as the strongest correlates of heart disease status among the original features. The engineered HRSI is expected to be a strong predictor as it combines both MaxHR and Age into a single normalised ratio.

4.5 Part 1 — Risk Classifier

4.5.1 Model 1: Random Forest

Listing 1: Random Forest configuration.

```
RandomForestClassifier(
    n_estimators = 300,
    max_depth    = None,    # fully grown trees
    class_weight = 'balanced',
```

```
random_state = 42,  
n_jobs       = -1  
)
```

`class_weight='balanced'` reweights each sample inversely proportional to class frequency, compensating for the underrepresented Medium class.

4.5.2 Model 2: XGBoost

Listing 2: XGBoost configuration.

```
XGBClassifier(  
    n_estimators = 400,  
    max_depth    = 6,  
    learning_rate = 0.05,  
    objective     = 'multi:softprob',  
    num_class     = 3,  
    eval_metric   = 'mlogloss',  
    random_state  = 42,  
    n_jobs        = -1  
)
```

4.5.3 Evaluation Metrics

Three metrics were used for comparison:

- **Accuracy:** Overall proportion of correctly classified samples.
- **F1-macro:** Unweighted mean of per-class F1-scores. Selected as the primary selection metric because it weights all three classes equally, penalising poor recall on the minority Medium class.
- **AUC-OVR:** Area Under the ROC Curve using a One-vs-Rest multi-class strategy, macro-averaged. Measures discriminability regardless of classification threshold.

4.6 Part 2 — Recommender and Survival Model

4.6.1 Clinical Routing

Before the recommender runs, a deterministic routing layer assigns each patient to a treatment branch based on:

$$\text{Branch} = \begin{cases} \text{SBRT} & \text{if RiskCategory} = \text{High} \wedge \text{has_arrhythmia} = \text{True} \\ \text{Medication} & \text{otherwise} \end{cases} \quad (1)$$

BED formula validation. For SBRT patients, the recommended dose of 25 Gy in a single fraction is validated using the Biologically Effective Dose formula (Joiner and van der Kogel, 2009):

$$\text{BED} = D \left(1 + \frac{d}{\alpha/\beta} \right) \quad (2)$$

where $D = 25$ Gy (total dose), $d = 25$ Gy (dose per fraction), and $\alpha/\beta = 10$ Gy for cardiac tissue. This yields:

$$\text{BED} = 25 \times \left(1 + \frac{25}{10} \right) = 25 \times 3.5 = 87.5 \text{ Gy} \quad (3)$$

This value falls within the published clinical window for cardiac radioablation established by Cuculich et al. (2017).

GRACE score calibration. For medication-path patients, the GRACE Risk Score (Eagle et al., 2004) is computed from age, heart rate, and systolic blood pressure using the published lookup tables, and medication intensity is mapped to one of three levels: Standard (GRACE <110), Intensified (GRACE 110–140), or Maximal (GRACE >140).

4.6.2 Intervention Recommender (XGBoost)

A second XGBoost classifier predicts the intervention label from the set $\{0 = \text{Medication-Standard}, 1 = \text{Medication-Intensified}, 2 = \text{Medication-Maximal}, 3 = \text{SBRT}\}$ with configuration: `n_estimators=300, max_depth=5, random_state=42`.

4.6.3 Survival Model

Two complementary survival models are fitted:

- **Kaplan–Meier Fitter:** Non-parametric survival curves stratified by treatment branch (SBRT vs. Medication), with 95% confidence bands. A log-rank test compares the two curves.
- **Cox Proportional Hazards:** Semi-parametric regression with covariates `{age, risk_encoded, grace_score, bed_gy}`. Produces per-patient two-year survival probabilities used in counterfactual output.

5 Results and Discussion

5.1 Part 1: Risk Classifier Results

5.1.1 Model Comparison

Table 3 presents the test-set performance of the two trained classifiers.

Table 3: Part 1 classifier performance on the held-out 20% test set.

Metric	Random Forest	XGBoost	Δ	Winner
Accuracy	0.9614	0.9579	+0.0035	Random Forest
F1-macro	0.9321	0.9251	+0.0070	Random Forest
AUC-OVR	0.9880	0.9853	+0.0027	Random Forest

Random Forest outperforms XGBoost on all three metrics. The performance gap is largest on F1-macro (+0.0070), which is the primary selection metric due to the minority Medium class.

5.1.2 Winner Rationale

The **Random Forest** is selected as the Part 1 model for the following reasons:

1. **Dataset scale.** At approximately 1,400 training samples, the dataset falls in the range where Random Forest and XGBoost perform comparably, but gradient boosting’s sequential error-correction mechanism introduces additional variance at small sample sizes. Random Forest’s parallel ensemble averaging is more stable in this regime.
2. **Class imbalance.** The `class_weight=‘balanced’` parameter in Random Forest redistributes sample weight per tree, providing fine-grained compensation for the underrepresented Medium class (8.4% of samples). XGBoost’s `scale_pos_weight` parameter is designed for binary tasks and does not directly map to the three-class scenario without additional post-processing.
3. **F1-macro superiority.** Because F1-macro weights all classes equally, the Random Forest’s advantage on this metric directly implies better recall on the minority Medium class — the clinically critical outcome where misclassification could result in under-treatment.

5.1.3 Feature Importance

The top 15 features by mean decrease in impurity from the winning Random Forest are:

Table 4: Top 10 feature importances from the winning Random Forest (indicative ranking).

Rank	Feature	Clinical Significance
1	Oldpeak (ST depression)	Direct marker of myocardial ischaemia
2	MaxHR	Chronotropic competence
3	HeartRateStressIndex	Engineered feature (validates design)
4	Age	Primary risk factor
5	ChestPainType_ASY	Asymptomatic ischaemia
6	ST_Slope_Flat	Exercise ECG abnormality
7	ExerciseAngina_Y	Exertional ischaemia
8	ST_Slope_Up	Protective ECG pattern
9	Cholesterol	Lipid burden
10	RestingBP	Chronic pressure load

Notably, the engineered `HeartRateStressIndex` ranks in the top three, validating the feature engineering decision to include it.

5.2 Part 2: Recommender and Survival Results

5.2.1 Intervention Recommender Results

Table 5: Part 2 XGBoost intervention recommender performance.

Metric	Value	Threshold	Status
Accuracy	1.00	≥ 0.75	Pass
F1-macro	1.00	≥ 0.75	Pass

The perfect scores require explanation: intervention labels are *deterministically derived* from clinical routing rules (RiskCategory + arrhythmia flag + GRACE score threshold). Because the routing rules are exact functions of features the model can observe, XGBoost with `max_depth=5` and 300 trees learns these decision boundaries without error. This is expected behaviour in the proof-of-concept setting. In a production deployment with real MIMIC-IV labels drawn from noisy clinical records, the accuracy would be substantially lower and more informative.

5.2.2 Survival Model Results

Table 6: Survival model performance.

Model	Metric	Value	Status
Cox PH	Concordance C-index	0.85	≥ 0.65 (Pass)
Log-rank test	p -value	< 0.05	Significant group difference

A C-index of 0.85 indicates strong discriminative ability: the Cox model correctly ranks the relative survival risk of 85% of patient pairs. The significant log-rank p -value confirms that the Kaplan–Meier survival curves for the SBRT and Medication branches differ significantly.

5.3 End-to-End Example Cases

Table 7 traces four representative patients through the complete pipeline.

Table 7: End-to-end pipeline results for four example patients.

Case	Patient	Part 1 Output	Recommendation	2-Year Survival
A	42 y/o M, healthy vitals	Low, 91%	Lifestyle + low-dose statin	95% $\rightarrow$ 97%
B	58 y/o M, ST abnormality	Medium, 83%	Beta-blocker + statin + aspirin	81% $\rightarrow$ 92%
C	67 y/o M, exercise angina, no arrhythmia	High, 88%	Aggressive multi-drug regimen	62% $\rightarrow$ 84%
D	71 y/o M, refractory VT	High, 94%	SBRT 25 Gy (BED = 87.5 Gy)	58% $\rightarrow$ 81%

These cases cover all four routing branches of the pipeline (Low, Medium, High without arrhythmia, High with arrhythmia) and demonstrate clinically plausible survival improvements that scale with baseline risk and intervention intensity.

5.4 Discussion

The CardioSurv pipeline achieves strong classification performance on Part 1, with Random Forest obtaining an F1-macro of 0.9321 and AUC-OVR of 0.9880 on the held-out test set. These results are consistent with findings from Mohan et al. (2019), who reported that ensemble methods consistently outperform single classifiers on clinical heart disease datasets. The marginal advantage of Random Forest over XGBoost at this dataset scale ($\approx 1,400$ samples) is also consistent with the literature: gradient boosting typically achieves larger gains over bagging ensembles at dataset sizes above 10,000 samples where iterative residual corrections provide compounding benefit.

The engineered `HeartRateStressIndex` ranking in the top three most important features validates the feature engineering step and demonstrates that normalising MaxHR by age-predicted maximum provides information beyond either feature alone. This is clinically interpretable: chronotropic competence — the ability to achieve near-predicted maximum heart rate under stress — is a recognised marker of cardiovascular fitness and ischaemic risk.

The Part 2 component demonstrates a complete clinical decision support workflow, from risk-stratified routing through validated treatment recommendation to counterfactual survival estimation. The integration of established clinical scoring tools (GRACE, BED) directly into the ML pipeline represents a principled approach to domain knowledge incorporation that improves both clinical plausibility and model trustworthiness.

6 Conclusion

This project developed and deployed CardioSurv, an end-to-end machine learning pipeline for cardiovascular risk triage that addresses the clinical challenge of identifying high-risk patients from routine, non-invasive vital sign measurements.

Key findings:

- Random Forest outperformed XGBoost on the Part 1 risk classification task at the dataset scale used ($\approx 1,400$ samples), achieving F1-macro of 0.9321 versus 0.9251.
- Feature engineering — particularly the Heart Rate Stress Index — contributed meaningfully to predictive performance, ranking third in feature importance.
- The two-stage pipeline (risk classification $\rightarrow$ treatment routing $\rightarrow$ survival estimation) successfully demonstrated a complete clinical decision support workflow for all four routing branches.

- The Cox PH survival model achieved a C-index of 0.85, indicating strong discriminative ability within the proof-of-concept setting.

Limitations:

- The survival component is based on synthetic simulation due to MIMIC-IV access constraints. Results from this component are proof-of-concept only and not clinically validated.
- The merged dataset of 1,221 rows is relatively small for deep learning approaches. Larger datasets would enable exploration of LSTM or Transformer architectures for temporal risk prediction.
- External validation on a held-out site-specific cohort was not performed, so generalisation to populations different from the training distribution is unknown.
- The XGBoost recommender achieves artificially perfect scores because intervention labels are deterministically derived; real-world deployment would require probabilistic labels from clinical records.

Future work:

- Integration with real MIMIC-IV longitudinal data to train and validate the survival component on observed patient outcomes.
- Hyperparameter optimisation via Optuna or Bayesian search for both models.
- Addition of explainability components (SHAP values) to support clinical interpretability.
- Evaluation of deep learning architectures (DeepSurv, TabNet) on larger data.
- Prospective clinical pilot study to validate risk categorisation against cardiologist assessment.

7 References

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Appendix A: Roles and Contributions

Table 8: Team member roles and contributions.

Member	Role	Responsibilities and Contributions
Ang Jing Ru	Feature Engineer	Repository setup, raw data download, schema documentation, dataset merging (<code>merge.py</code>), data cleaning (<code>clean.py</code>), feature engineering (<code>engineering.py</code>), and production of <code>features.csv</code> . Implemented <code>age_bin()</code> , <code>bp_risk_level()</code> , <code>hr_stress_index()</code> , and <code>derive_risk_label()</code> functions with unit tests.
Bougacha Mohamed	ML Engineer	Designed and trained the Part 1 risk classifier (Random Forest and XGBoost comparison, <code>02_part1_modeling.ipynb</code>). Designed and trained the Part 2 intervention recommender and survival model (XGBoost + Kaplan–Meier + Cox PH, <code>03_part2_modeling.ipynb</code>). Implemented the BED formula and GRACE score clinical helpers. Serialised all model bundles (<code>.pkl</code> files).
Law Zi Ying	Frontend Developer	Designed and built the complete four-page frontend interface (<code>index.html</code> , <code>predict.html</code> , <code>results.html</code> , <code>history.html</code>), CSS variable system, and JavaScript API layer (<code>api.js</code>). Implemented client-side validation, Chart.js survival bar chart, and accessibility features. FastAPI scaffold setup.
Chiluba Chifunilo	Backend Engineer	Implemented database models (patients, predictions, recommendations tables via SQLAlchemy), API endpoint integration, seed data, production deployment to Render, error handling, rate limiting, and deployment documentation (<code>deployment.md</code>).
Tan Guan Han	EDA & Evaluation	Conducted exploratory data analysis (<code>01_eda.ipynb</code>), generated 5 publication figures. Implemented evaluation metrics (<code>metrics.py</code>) and plotting functions (<code>plots.py</code>). Wrote unit tests for BED formula (<code>test_bed.py</code>) and GRACE score (<code>test_grace.py</code>).

Appendix B: Model Configuration Summary

Table 9: Summary of all models trained in the CardioSurv pipeline.

Model	Stage	Key Hyperparameters
Random Forest	Part 1	<code>n_estimators=300</code> , <code>max_depth=None</code> , <code>class_weight='balanced'</code> , <code>random_state=42</code>
XGBoost	Part 1	<code>n_estimators=400</code> , <code>max_depth=6</code> , <code>learning_rate=0.05</code> , <code>objective='multi:softprob'</code> , <code>num_class=3</code> , <code>eval_metric='mlogloss'</code>
XGBoost	Part 2 Rec-ommender	<code>n_estimators=300</code> , <code>max_depth=5</code> , <code>objective='multi:softprob'</code> , <code>num_class=4</code> , <code>random_state=42</code>
Kaplan–Meier	Survival	Non-parametric; stratified by treatment branch (SBRT vs. Medication); 95% CI bands
Cox PH	Survival	Covariates: <code>age</code> , <code>risk_encoded</code> , <code>grace_score</code> , <code>bed_gy</code> ; concordance C-index = 0.85